5CM510 – Network System Development

Lab: 5G and Beyond Using Cisco Packet Tracer

Detailed step-by-step lab instruction material and built around the Packet Tracer Cell Tower device

Mode	Focus	Deliverable
Guided lab	Cellular access, edge services, smart-city style traffic	Working Packet Tracer file + short analysis

Recommended duration: 120–150 minutes | **Software:** Cisco Packet Tracer | **Difficulty:** Intermediate

Important: Packet Tracer is used here as a teaching platform for cellular connectivity and service orchestration. It can demonstrate logical attachment, IP reachability, edge/server placement, and multi-device access, but it does not behave like a standards-accurate 5G NR or 6G simulator.

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1. Lab overview and learning outcomes

In this lab you will build a small cellular-to-edge network in Cisco Packet Tracer using the Cell Tower device, a Central Office Server, a router, a switch, an edge application server, and one or more mobile clients. The purpose is to create a realistic classroom exercise that lets you discuss 5G ideas—higher data rates, lower latency, massive device connectivity, mission-critical services, and edge computing—while remaining honest about what Packet Tracer can and cannot model.

The completed topology represents a simplified smart-city / smart-campus deployment. A mobile user reaches an edge service over a simulated cellular access network. You will then extend the topology to discuss density, service separation, edge-vs-cloud thinking, and 6G design directions.

By the end of the lab, students should be able to:

- Explain why 5G improves on 4G for high-capacity, low-latency, and massive-connectivity use cases.
- Build a Packet Tracer topology that uses the Cell Tower device and Central Office Server for cellular-style access.
- Configure IP addressing, routing, and application services so that a smartphone can reach an edge server through the cellular path.
- Verify connectivity with ping, browser access, and simulation-mode packet inspection.
- Relate lab observations to lecture themes such as smart cities, critical services, dense user scenarios, and the move from 5G toward 6G.
- Evaluate the limitations of an educational simulator when discussing advanced topics such as Massive MIMO, network slicing, THz bands, RIS, and four-tier 6G networks.

2. Background and concept-to-lab mapping

Lecture 18 describes the move from analogue 1G to digital 2G, packet-switched 3G, all-IP 4G, and then 5G with increased data rates, critical services, and support for billions of devices. The lecture also highlights 5G access scenarios such as device-to-device communication, machine-type communication, and dense-user deployments, alongside technologies such as Filtered OFDM, SCMA, polar codes, and Massive MIMO. Finally, it introduces a 6G vision with space-air-ground-underwater tiers, mmWave/THz communications, reconfigurable intelligent surfaces, and intelligent software-defined networks.

Use the table below to connect the lecture ideas to this Packet Tracer activity.

Lecture concept	What Packet Tracer can show	How you will address it in the lab
Higher 5G capacity and broadband access	Logical mobile attachment and end-to-end reachability	Use a smartphone through the Cell Tower to access an edge server.
Low latency / mission-critical traffic	Shorter logical path to an edge service	Host the service close to the radio side and compare it conceptually with a cloud extension.
Massive connectivity / smart city devices	Multiple clients or IoT nodes attaching to one access area	Duplicate mobile clients or optional IoT endpoints and verify simultaneous reachability.
Massive MIMO and dense areas	Only conceptual discussion; not actual radio beamforming	Discuss why real 5G uses massive antenna arrays for stadiums and urban centres.
Network slicing	Approximation with VLANs / ACLs / service separation	Optional extension: separate critical, sensor, and broadband traffic.
6G four-tier architecture	Not directly simulated	Use a design extension to map space, air, ground, and underwater tiers beyond the Packet Tracer build.

3. Simulation scope, assumptions, and limitations

Teaching stance: Use the Packet Tracer Cell Tower as a cellular-access abstraction rather than a literal 5G NR base station. This lab focuses on architecture, addressing, reachability, service placement, and teaching discussion.

- The Cell Tower device in Packet Tracer provides the access node used in this exercise. In current Packet Tracer device help it is shown with one slot and support for coaxial and 3G/4G connectivity, so you should treat it as a generic cellular access element rather than a standards-accurate 5G gNodeB.
- Packet Tracer documentation for end-device 5G modules states that the tool models only the antenna-level capability and does not model actual 5G technologies. Therefore, concepts such as numerology, NR scheduling, beam management, HARQ behaviour, QoS flows, and radio propagation are outside the scope of this lab.
- Massive MIMO, F-OFDM, SCMA, polar coding, mmWave, THz communications, RIS, and space-air-ground-underwater networking are discussed conceptually in the analysis sections rather than reproduced as physical-layer simulations.
- Menu names vary slightly between Packet Tracer versions. If a configuration tab is labelled differently in your installation, use the nearest equivalent option—for example, 3G/4G Cell, Cellular, Provider, or Global Settings.

4. Topology, device list, and addressing plan

4.1 Reference topology

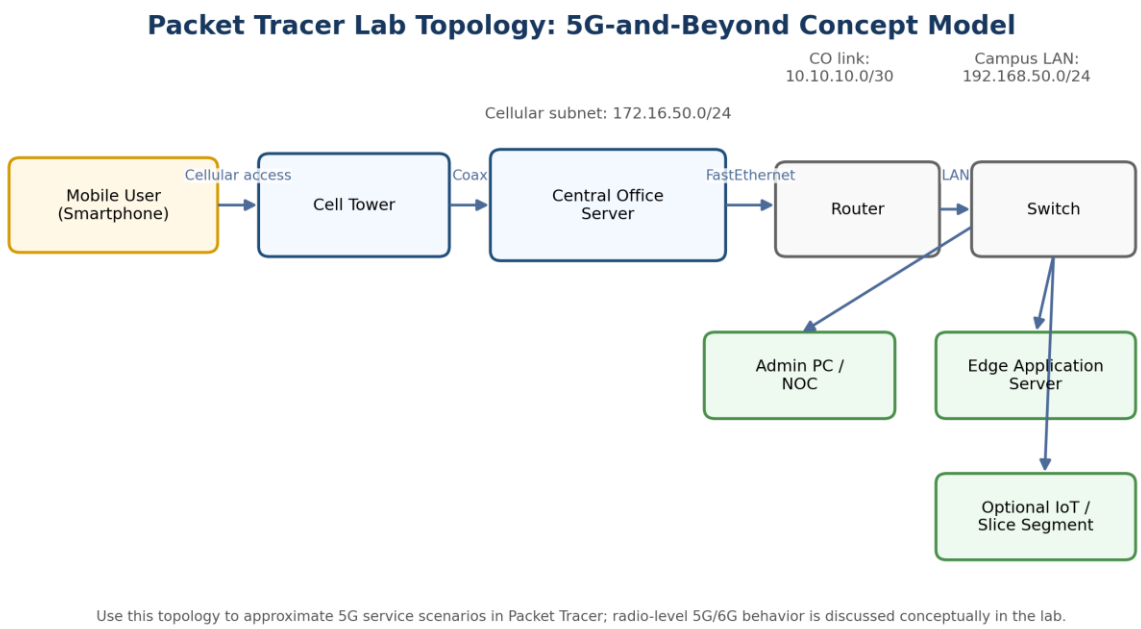

Figure 1. Recommended logical topology for the lab.

4.2 Required devices

Device	Suggested count	Role in the lab	Notes
Cell Tower	1	Cellular access point	Use the Packet Tracer Cell Tower device. If the cellular module is not already present, insert the supported module from the Physical tab.
Central Office Server	1	Back-end cellular / provider node	Connect to the tower using a coaxial link and to the router using FastEthernet.
Router with two Gigabit/FastEthernet interfaces	1	Routed hand-off to the campus / edge LAN	A 2911 or similar Packet Tracer router works well.
2960 switch or equivalent	1	Access switch for server and PC	Any Layer-2 access switch is acceptable.
Server-PT	1	Edge application server	Enable HTTP and optionally DNS.
PC-PT or Laptop-PT	1	Admin / NOC node for testing	Use this for console access, ping, and browser tests.
Smartphone	1–4	Mobile user(s)	One is mandatory; extra clients support density and smart-city discussion.

4.3 Addressing plan

Segment / Device	Interface	IP address	Subnet mask	Default gateway
Router to Central Office link	Router G0/0	10.10.10.1	255.255.255.252	N/A
Central Office Server	FastEthernet0	10.10.10.2	255.255.255.252	10.10.10.1
Cellular subnet	Provider / 3G-4G network	172.16.50.0/24	255.255.255.0	172.16.50.1
Edge LAN gateway	Router G0/1	192.168.50.1	255.255.255.0	N/A
Edge application server	FastEthernet0	192.168.50.10	255.255.255.0	192.168.50.1
Admin PC	FastEthernet0	192.168.50.20	255.255.255.0	192.168.50.1
Smartphone	Cellular	DHCP / auto from 172.16.50.0/24	255.255.255.0	172.16.50.1

Addressing rule: You may change the addressing scheme to fit your module conventions, but keep three separate logical networks: (1) the Central Office–router hand-off, (2) the cellular client subnet, and (3) the edge LAN.

5. Part A – Build the base topology

Create a new Packet Tracer file and save it immediately with a clear name such as "5CM510_Lab18_5G_Beyond_<studentID>.pkt". Throughout the build, save frequently.

Step 1. Place the following devices in the logical workspace: one Cell Tower, one Central Office Server, one router, one switch, one Server-PT, one PC-PT, and one Smartphone.

Step 2. Arrange the devices from left to right to reflect the service path: Smartphone → Cell Tower → Central Office Server → Router → Switch → Server/PC. This makes later troubleshooting much easier.

Step 3. Use a coaxial cable to connect the Cell Tower to one coaxial port on the Central Office Server.

Step 4. Use a copper straight-through cable to connect the Central Office Server FastEthernet port to the router interface that will face the provider side.

Step 5. Use a copper straight-through cable to connect the router LAN interface to the switch.

Step 6. Connect the Server-PT and the PC-PT to the switch using copper straight-through cables.

Step 7. Rename the devices so the workspace is readable. Suggested names: CELL-TOWER, CO-SERVER, RAN-EDGE, EDGE-SW, EDGE-SRV, NOC-PC, and USER-PHONE.

If your Packet Tracer workspace also shows a physical view, you can ignore it for this lab unless your instructor specifically asks for physical placement. The logic of the lab is easier to follow in the logical view.

6. Part B – Configure the Central Office Server and Cell Tower

6.1 Configure the Central Office Server

Step 1. Click CO-SERVER and open the Config tab.

Step 2. Select the FastEthernet interface and assign the static IP address 10.10.10.2 with subnet mask 255.255.255.252.

Step 3. Set the default gateway of CO-SERVER to 10.10.10.1 so that traffic can leave the provider node toward the router.

Step 4. Locate the cellular / 3G/4G configuration area on the Central Office Server. In many Packet Tracer builds this appears as a dedicated configuration page for the provider-side cellular network.

Step 5. Create or enable a provider profile named UoD-5G-Lab (or another clearly identifiable provider name approved by your instructor).

Step 6. Configure the cellular addressing so the phone clients will attach to subnet 172.16.50.0/24. If your version asks for a gateway or provider address, use 172.16.50.1.

Step 7. If the server provides address pooling options, define a client range such as 172.16.50.100 to 172.16.50.150.

6.2 Configure the Cell Tower

Step 1. Click CELL-TOWER. First check whether the device is already operational and whether a cellular-capable module is present.

Step 2. If the tower requires a module installation, open the Physical tab, power off the device, insert the supported 3G/4G module into the slot, and power the device back on.

Step 3. Return to the Config tab and match the provider settings to the Central Office Server. Use the same provider name: UoD-5G-Lab.

Step 4. Verify that the coaxial link between the tower and the Central Office Server is up. If the link is down, recheck the cable type and the physical module state.

Step 5. Wait a few seconds or use Fast Forward Time so Packet Tracer can converge.

Common issue: A common failure point is mismatched provider names between the tower, the central office settings, and the smartphone. Use exactly the same spelling and spacing everywhere.

7. Part C – Configure the edge LAN, routing, and services

7.1 Router configuration

Open the router CLI and apply the following configuration. Adjust interface names if your router model uses different numbering.

```
enable
configure terminal
hostname RAN-EDGE
interface g0/0
 description Link_to_Central_Office
 ip address 10.10.10.1 255.255.255.252
 no shutdown
exit
interface g0/1
 description Edge_LAN
 ip address 192.168.50.1 255.255.255.0
 no shutdown
exit
ip route 172.16.50.0 255.255.255.0 10.10.10.2
end
write memory
```

7.2 Configure the Edge Application Server

Step 1. Open EDGE-SRV and go to Desktop > IP Configuration (or Config > FastEthernet, depending on version).

Step 2. Assign IP address 192.168.50.10, subnet mask 255.255.255.0, default gateway 192.168.50.1, and DNS 192.168.50.10.

Step 3. Go to Services and turn on HTTP.

Step 4. Edit the default web page so it clearly identifies the server, for example: '5CM510 Edge Dashboard – Smart City / Smart Campus Services'.

Step 5. Turn on DNS and create an A record such as edge.uod5glab.local pointing to 192.168.50.10.

7.3 Configure the Admin / NOC PC

Step 1. Open NOC-PC and assign IP address 192.168.50.20, subnet mask 255.255.255.0, default gateway 192.168.50.1, and DNS 192.168.50.10.

Step 2. From the PC Command Prompt, ping 192.168.50.10 and then ping 10.10.10.2. Both should succeed before you move on.

Step 3. Use the PC web browser to open http://192.168.50.10 and then http://edge.uod5glab.local to verify both HTTP and DNS functionality.

8. Part D – Attach a smartphone through the cellular network

This is the key section of the lab. The smartphone must attach through the Cell Tower rather than through a Wi-Fi network. Packet Tracer versions differ slightly, but the logic remains the same: select the cellular provider, let the phone register, verify it receives an address from the cellular subnet, and test reachability to the edge server.

Step 1. Open USER-PHONE and go to the Config tab.

Step 2. Find the cellular settings. The menu may be labelled 3G/4G Cell, Cellular, or similar.

Step 3. Set the provider name to UoD-5G-Lab. If the phone has an option to search for Wi-Fi by default, change it so that the phone uses the cellular network instead.

Step 4. Wait for the device to register. If needed, click Fast Forward Time several times.

Step 5. Check the smartphone IP settings. The phone should receive an address from 172.16.50.0/24, for example 172.16.50.100.

Step 6. Open the smartphone command prompt and test the following pings in order:

```
ping 172.16.50.1
ping 10.10.10.1
ping 192.168.50.10
ping 192.168.50.20
```

Step 7. Open the smartphone web browser and browse to http://192.168.50.10. Then browse to http://edge.uod5glab.local.

Step 8. Take a screenshot showing successful browser access from the smartphone through the cellular path. This is one of your key pieces of evidence.

If the smartphone does not attach, confirm these four items in order: (1) the tower and central office are linked by coax, (2) the provider name matches on all devices, (3) the router has a route back to the cellular subnet, and (4) the phone is set to cellular rather than Wi-Fi.

9. Part E – Verification, testing, and evidence capture

Use the checks below to validate the build and to create evidence for your lab submission.

- From NOC-PC, ping the router LAN interface, the Central Office Server, and the edge server.
- From USER-PHONE, ping the cellular gateway, the router-facing provider link, and the edge server.
- Access the edge dashboard from both the PC and the smartphone using HTTP.
- Use Simulation Mode to capture at least one smartphone-to-server transaction. Observe the path through Cell Tower → Central Office Server → Router → Switch → Server.
- Document the smartphone IP address and explain which subnet assigned it.
- Save the .pkt file after successful testing.

Recommended evidence table

Evidence item	What to capture	Why it matters
Topology screenshot	Logical topology with all device names visible	Shows that the required architecture was built correctly.
PC connectivity test	Successful ping and browser access from NOC-PC	Confirms edge LAN and services are working.
Smartphone connectivity test	Phone IP address plus successful ping to 192.168.50.10	Confirms the cellular attachment path.
Smartphone browser view	HTTP page loaded from edge.uod5glab.local or 192.168.50.10	Demonstrates service delivery over the cellular path.
Simulation screenshot	One captured packet path through the tower and central office	Links practical work to network-layer understanding.

10. Part F – Extend toward 5G service classes and beyond-5G ideas

10.1 Extension A – Approximate eMBB, URLLC, and mMTC

- eMBB-style scenario: Keep the smartphone-to-edge dashboard workflow and describe it as a simplified high-throughput mobile broadband service.
- URLLC-style scenario: Add a second HTTP page or a second server labelled 'Critical Service' and explain why placing the service at the edge shortens the path for time-sensitive traffic.
- mMTC-style scenario: Add extra smartphones or optional IoT devices so several endpoints are attached to the same provider area. Verify that each receives connectivity.

10.2 Extension B – Service separation as a network slicing approximation

True 5G network slicing is not implemented in Packet Tracer, but you can approximate the idea by separating services into VLANs or different IP subnets. A good classroom extension is to create three logical groups:

- VLAN 10 – Critical service / healthcare / emergency traffic
- VLAN 20 – Smart-city sensors and telemetry
- VLAN 30 – General mobile broadband users

After creating the VLANs, use router subinterfaces or separate routed interfaces, then restrict or prioritize access using ACLs or simple QoS features if your course has already covered them.

10.3 Extension C – Dense user discussion and Massive MIMO reflection

Duplicate the smartphone three times so four users attach to the same provider area. Run several pings or browser requests. Packet Tracer will not show beamforming or spectral efficiency gains, but this extension supports a discussion of why real 5G relies on dense small cells and Massive MIMO in stadiums, urban centres, and transport hubs.

10.4 Extension D – Beyond 5G / 6G design worksheet

After completing the Packet Tracer build, answer the following design prompts:

- How would you extend your topology from a ground-only cellular system to a four-tier 6G vision with space, air, ground, and underwater segments?
- Which parts of the system would benefit most from AI-driven network intelligence or adaptive slicing?
- Where would mmWave or THz links be most useful - access, backhaul, or both?
- What role could reconfigurable intelligent surfaces play in dense indoor or urban deployments?
- Which use case best fits your completed lab topology: smart city, healthcare, transport, or immersive mixed reality?

11. Troubleshooting guide

Symptom	Likely cause	Fix
Smartphone gets no IP address	Provider mismatch or phone still prefers Wi-Fi	Set the same provider name everywhere and force the phone to use cellular.
Phone can ping gateway but not the edge server	Missing static route on router	Add ip route 172.16.50.0 255.255.255.0 10.10.10.2 and verify interface status.
Tower appears down	Module missing or device powered off	Install the supported module from the Physical tab and power the device on.
PC can reach the server but phone browser fails	HTTP or DNS service is off	Turn on HTTP/DNS on EDGE-SRV and verify the DNS A record.
Intermittent Packet Tracer behaviour after changes	Convergence delay	Use Fast Forward Time and retest.

12. Submission checklist, reflection questions, and marking guide

12.1 Submission checklist

- Saved Packet Tracer file (.pkt) with clear device names
- At least one screenshot of the completed topology
- At least one screenshot of successful smartphone access through the cellular path
- Short answer section addressing the reflection questions below
- Any optional extension work clearly labelled

12.2 Reflection questions

1. Which limitation of 4G is most clearly addressed by the architecture you built in this lab?

2. Why is hosting the application on the edge server a useful way to discuss low-latency 5G services?

3. Which part of the lab best represents smart-city or massive-machine-type communication, and why?

4. Why can this Packet Tracer exercise support a discussion of 5G and 6G even though it does not simulate actual NR or 6G radio behaviour?

5. Name two 6G enablers from the lecture that cannot be reproduced directly in Packet Tracer and explain how you would discuss them instead.

12.3 Simple marking guide (example)

Criterion	Marks	Expectation
Correct topology and cabling	20	All mandatory devices are present and connected correctly.
Addressing and routing	20	Interfaces, gateway values, and route to cellular subnet are correct.
Cellular attachment	20	Smartphone successfully joins the provider network and receives reachability.
Application / browser test	15	Smartphone accesses the edge server over the cellular path.
Analysis and reflection	15	Responses clearly connect the build to 5G and beyond concepts.
Presentation and evidence	10	Screenshots, file naming, and documentation are complete and readable.

References used to prepare this lab

- University of Derby, 5CM510 – Network System Development, Lecture 18: 5G towards 6G Networks.
- Cisco Packet Tracer device help for the Cell Tower and Central Office Server device family.
- Cisco Packet Tracer end-device help for cellular-capable host modules, including the note that 5G capability is modeled at the interface/antenna level rather than as a full 5G technology simulation.
- Cisco Community guidance on configuring Central Office Server addressing, the cellular network, and handset attachment behaviour in Packet Tracer.